

KHALIL BOU KHALIL

Junior AI Engineer | LLM Applications, AI Agents, RAG and Machine Learning

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PROFILE

AI Master's candidate at CNAM Paris building end-to-end AI applications: LLM agents, RAG pipelines, FastAPI backends and machine learning models. Currently developing CTCARCH's internal multi-model AI platform. **Seeking a full-time AI Engineer position in France, available from October 2026.**

EDUCATION

- CNAM Paris – Master 1 & Master 2, Artificial Intelligence for Connected Industries (AI4CI) · Paris, France2024 – 2026
- Lebanese International University (LIU) – BSc in Computer Science · Koura, Lebanon2021 – 2024

PROFESSIONAL EXPERIENCE

- CTCARCH Pty Ltd – AI Engineer Intern · Remote, company based in Sydney, AustraliaJul 2026 – Present
 - Analyse workflows across architecture, construction, estimating and operations to identify practical AI use cases.
 - Architect and develop the foundation of an internal multi-model AI platform that centralises company knowledge and delivers department-specific assistants to around 12 employees (Python, FastAPI, Next.js, PostgreSQL, Docker).
 - Build the RAG and document-processing pipeline: PDF, Word and Excel ingestion, parsing, chunking, embeddings and pgvector search, powering document question answering with project context.
 - Implement the orchestration flow – task classification, retrieval, prompt construction, automatic model routing across OpenAI, Anthropic and Google APIs, human review – with routing driven by task type, quality, speed, cost and privacy.
 - Specify the governance layer under development: role-based access control, permissions, employee quotas, token and cost tracking, audit logs and an administrator dashboard.
- Safo Systems – AI Agents Developer Intern · Tripoli, LebanonJul 2025 – Sep 2025
 - Developed Python multi-agent workflows automating document generation, with drafting, validation and human approval.
 - Applied LLM APIs, prompt engineering, tool calling and backend integration, and refined agent behaviour for reliability.
- MA Tech – Full-Stack Developer Intern · Minieh-Dannieh, LebanonJun 2024 – Sep 2024
 - Contributed to full-stack React.js and Node.js applications: frontend interfaces, dashboards, backend services and APIs.
 - Participated in testing, debugging, performance improvements and Git-based development workflows.

SELECTED TECHNICAL PROJECTS

- DocuRAG – Document Question-Answering Platform | Python · FastAPI · React · FAISS · SQLAlchemy · Docker
 - Built a full-stack RAG application combining FAISS vector search and BM25 into a hybrid retrieval pipeline over sentence-transformer embeddings, returning answers with source citations.
 - Added JWT authentication, SQLAlchemy persistence, rate limiting and Docker packaging over Ollama or LLM APIs.
- GenAI Multi-Agent Automation System | Python · LLM APIs · FastAPI · Streamlit
 - Designed a three-agent workflow for data extraction, memo generation and human-in-the-loop validation in Python.
 - Exposed the workflow through a FastAPI and Streamlit interface for non-technical review and approval.

TECHNICAL SKILLS

- Languages: Python, JavaScript, TypeScript, Java, SQL, Bash
- LLM Engineering: LLM APIs (OpenAI, Anthropic, Google), AI agents, RAG, prompt engineering, embeddings, vector search
- Machine Learning: PyTorch, TensorFlow, Scikit-learn, LSTM, model training and evaluation, data preprocessing
- Backend & Data: FastAPI, Flask, Node.js, REST APIs, authentication, RBAC, PostgreSQL, pgvector, FAISS, MongoDB
- Frontend & DevOps: React, Next.js, Docker, Git, GitHub, Linux

LANGUAGES

- Arabic – Native
- English – B2, professional working proficiency
- French – B1, intermediate